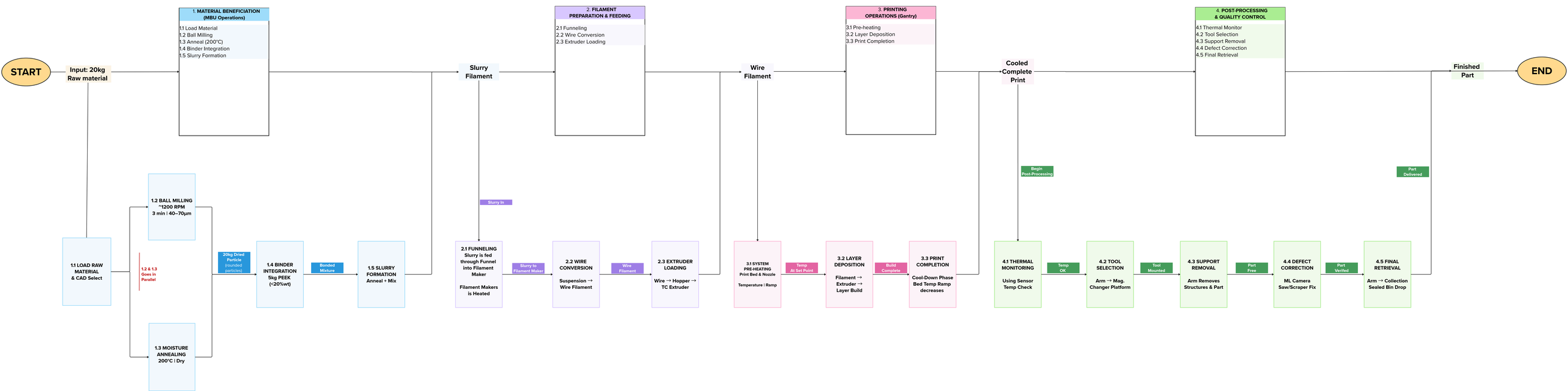
See an example

AERIS: Functional Flow Block Diagram (FFBD)



SUBSYSTEM DEVICES & SENSORS
All components interfacing with this function block

1. Dual Centrifuge Mixer & Sieve (**Device**)
(MBU Module Capacity: ⩾20kg)

2. Ball Mill Mechanism (**Device**)
(~1200 RPM | Output: 40–70µm)

3. Band Heater & Thermocouple Sensor (**Sensor**)
Target: 200°C & Moisture removal

4. Binder Intake Chamber (**Device**)
PEEK Pellets | ⩽20% total weight

5. Funnel System (Transfer Channel) (**Device**)
Slurry Feed Line | To Filament Maker

6. SWIR Spectroscopy Module (**Camera**)
Spectral Analysis

7. HD Camera System (MBU) (**Camera**)
High-Speed | Defect Detection

8. Pressure Sensor Array (**Sensor**)

SUBSYSTEM DEVICES & SENSORS
All components interfacing with this function

1. Funnel System For Slurry Transfer (**Device**)
Gravity/Pressurized | Feed into Maker

2. Heated Filament Maker (**Device**)
Conversion Unit | Temp-Controlled

3. Band Heater & Thermocouple Sensor (**Sensor**)
Filament Temp | Monitor (TC)

4. Tungsten Carbide Extruder (**Device**)
Anti-Agglomeration | Nozzle + Barrel

5. Hopper (Pellet Feed) (**Device**)
PEEK Pellets | Inlet Port

6. Stepper Motor & Planetary Gearbox (**Device**)
Filament Drive | Speed Control

7. Aluminum Extrusion Mount (**Device**)
Structural Extruder Housing

8. Extruder Screw for Feed Mechanism (**Device**)
Positive Feed | Anti-Slip

9. Sheet Metal Cover & Fan (**Device**)
Heat Dissipation | Air Cooling

10. Coupling for Motor-Screw (**Device**)
Mechanical Transmission

11. Cables Out for Power & Data
Stepper Power | Encoder Feedback

SUBSYSTEM DEVICES & SENSORS
All components interfacing with this function

1. 3D Printer Gantry System (**Device**)
XYZ Motion | Coordinate Control

2. Ceramic Build Plate (**Device**)
High-Temp Stable | Adhesion Surface

3. Anti-Agglomeration Extruder Nozzle (**Device**)
Tungsten Carbide Tip | Heated Barrel

4. Band Heaters for Nozzle & Barrel (**Device**)
Temp Setpoint | Zone Control

5. Thermocouples for Nozzle & Bed (**Sensor**)
Real-Time Temp Feedback | ±1°C

6. Vacuum Chamber for Print Enclosure (**Device**)
Oxidation Control | Inert Atmosphere

7. Crucible for melting material (**Device**)
High-Temp | Containment

8. HD Camera System (Gantry) (**Camera**)
Layer Inspection | Real-Time QC

9. High-Speed Camera Array (**Sensor**)
Motion Capture | Defect Detection

10. Ruggedized Interface Screen (**Device**)
Operator HMI | Status Display

11. Power System & Batteries
Peak: 3kW | Cont: 2.5kW

SUBSYSTEM DEVICES & SENSORS
All components interfacing with this function

1. Thermal Sensor Array (**Sensor**)
Print Chamber | Temp Monitoring

2. 5DOF Robotic Arm (**Device**)
Joint: Base/Shld/Elbow/Wrist | To control Yaw

3. Magnetic Tool Changer Platform (**Device**)
Auto End-Effectorl Swap System

4. End Effector Suite: Gripper (**Device**)
Saw | Scraper Support Tools

5. ML-Reinforced Camera System (**Sensor**)
Delamination & Adhesion Detection

6. Sealed Collection Bin (**Device**)
Final Part Deposit Area

7. High-Speed Camera for QC (**Sensor**)
ML Defect ID | Correction Trigger

8. Trigger System (Cooling → Arm) (**Sensor**)
Temp Threshold | Activation Logic

9. Ceramic Build Plate (**Device**)
Part Adhesion | Release Surface

10. Scraper Tool End Effector (**Device**)
Surface Finish | Burr Removal

11. Saw Tool End Effector (**Device**)
Support Cut | Overhang Trim

12. Ruggedized Interface Screen (**Device**)
Status Display | User Alert